

Grounding Stress-Test Matrix

Sanity probes separate answer accuracy, control robustness, and explanation grounding.

One-glance insight: a model can reject blank images yet still rationalize wrong or mismatched visual evidence.

1. Probe outcomes

Green = expected grounding behavior observed, red = contradictory support, gray = heuristic unclear.

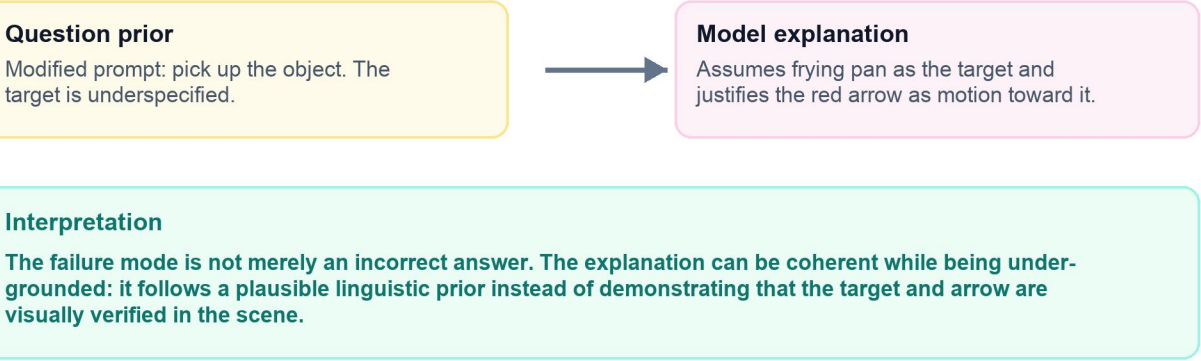
	Supports correct image Expected: support	Rejects wrong answer Expected: reject	Rejects blank image Expected: reject	Rejects shuffled image Expected: reject	Failure fingerprint Compact interpretation of the probe pattern.
Qwen2.5-VL-3B n = 20 probes per type Sanity source: spatial CoT	12/20 support 8/20 unclear	0/20 reject 16/20 contrad. 4/20 unclear	20/20 reject	3/20 reject 17/20 unclear	Sees absence, misses mismatch Detects blank inputs, but often accepts wrong answers or shuffled evidence.
DeepSeek-VL2 tiny n = 20 probes per type Sanity source: spatial CoT	0/20 support 20/20 unclear	3/20 reject 17/20 unclear	20/20 reject	20/20 reject	Rejects controls, weak positive grounding Flags blank and shuffled controls, but rarely phrases correct-image explanations as support.

expected behavior observed contradictory support heuristic unclear

2. Misleading-question anomaly

A qualitative stress case for ambiguity and insufficient image grounding.

source_index 408 image context



Takeaway: accuracy, control robustness, and explanation grounding are separable axes. Sanity evaluations are most useful when they reveal which axis fails.